

Infosys SP Role Interview Questions

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Coding / DSA

1. Write a program to find the first non-repeating character in a string.
2. Reverse a linked list iteratively and recursively.
3. Detect a cycle in a linked list and return the starting node.
4. Find the intersection point of two linked lists.
5. Check whether a binary tree is height balanced.
6. Find the lowest common ancestor of two nodes in a binary tree.
7. Print all root-to-leaf paths in a binary tree.
8. Convert a binary search tree to a sorted doubly linked list.
9. Find the kth smallest element in a BST.
10. Implement a stack using queues.
11. Implement a queue using stacks.
12. Find the next greater element for each element in an array.
13. Find the maximum sum subarray.
14. Find the longest substring without repeating characters.
15. Check whether two strings are anagrams.
16. Find all permutations of a string.
17. Rotate an array by k positions.
18. Merge two sorted arrays without using extra space.
19. Find the missing number in an array of size n containing 1 to n.
20. Find duplicate elements in an array without extra space.
21. Sort an array of 0s, 1s, and 2s.

22. Find the median of two sorted arrays.
23. Given an array, find the equilibrium index.
24. Find the longest increasing subsequence.
25. Implement binary search without recursion.
26. Explain and code quicksort.
27. Explain and code mergesort.
28. Find the shortest path in an unweighted graph.
29. Detect a cycle in a directed graph.
30. Count the number of islands in a matrix.
31. Find all pairs with a given sum.
32. Check whether a number is a power of two.
33. Reverse bits of an integer.
34. Find the maximum product subarray.
35. Given a matrix, rotate it by 90 degrees.

Programming / Logic

36. Difference between pass-by-value and pass-by-reference with example.
37. What happens internally when you create an object?
38. How does garbage collection work?
39. Explain method overloading vs overriding with code.
40. What is immutability? Why are strings immutable?
41. Difference between abstract class and interface.
42. Can a constructor be overridden?
43. Can we make an object without using new?
44. What happens if main method is declared final/static/private?
45. Explain shallow copy vs deep copy.
46. Why is multiple inheritance not supported in some languages?
47. What are lambda expressions and where do you use them?

- 48. Difference between checked and unchecked exceptions.
 - 49. What happens when an exception is not handled?
 - 50. Explain the difference between == and .equals().
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DBMS / SQL

- 51. Write a query to find the second highest salary.
 - 52. Difference between WHERE and HAVING.
 - 53. Difference between DELETE, TRUNCATE, and DROP.
 - 54. What is normalization? Why is it required?
 - 55. Explain ACID properties.
 - 56. What is an index? How does it improve performance?
 - 57. What are joins? Explain with examples.
 - 58. What is a deadlock in DBMS?
 - 59. Difference between clustered and non-clustered index.
 - 60. What is a transaction and rollback?
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Operating Systems

- 61. What is a deadlock? Necessary conditions?
 - 62. Difference between process and thread.
 - 63. What is context switching?
 - 64. Explain paging and segmentation.
 - 65. What is virtual memory?
 - 66. What is starvation?
 - 67. Explain scheduling algorithms.
 - 68. Difference between semaphore and mutex.
 - 69. What happens when a program is executed?
 - 70. Explain memory allocation techniques.
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Computer Networks

- 71. Difference between TCP and UDP.
 - 72. What happens when you enter a URL in a browser?
 - 73. Explain OSI model layers.
 - 74. Difference between HTTP and HTTPS.
 - 75. What is DNS?
 - 76. What is a socket?
 - 77. What is congestion control?
 - 78. Explain three-way handshake.
 - 79. Difference between IPv4 and IPv6.
 - 80. What is latency vs bandwidth?
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Projects / Scenario-Based

- 81. Explain your project architecture.
 - 82. What challenges did you face in your project?
 - 83. How did you optimize your project?
 - 84. How would you scale your application?
 - 85. How would you handle millions of user requests?
 - 86. What design changes would you make if traffic increases 10×?
 - 87. How do you debug production issues?
 - 88. What will you do if your code fails in production?
 - 89. Explain a real-time problem you solved using code.
 - 90. How do you ensure code quality?
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Behavioral / HR-Style (Still Asked in SP)

- 91. Why should we hire you for the SP role?
- 92. What makes you different from other candidates?
- 93. How do you handle tight deadlines?

94. Describe a situation where you failed.
95. Are you comfortable learning new technologies quickly?
96. How do you handle criticism?
97. Would you relocate if required?
98. Where do you see yourself in 5 years?
99. What will you do if assigned work outside your skill set?
100. Why Infosys SP and not other roles?



Real Candidate-Reported Infosys SP Interview Questions

Coding / Problem-Solving (from interviews & coding rounds)

1. *Graph Problem (Minimum Jumps on Circular Chairs)* – Given N chairs in a circle and jump lengths, find minimum jumps from X to Y or return -1.
2. Merge two sorted linked lists.
3. Binary tree related problems (varied DSA).
4. *Queue + Greedy* problem (medium – from online coding test).
5. DP + GCD problem (medium-hard – coding test).
6. DP + HashMap + Prime Factorization problem (hard – coding test).
7. Basic array coding problem (unspecified but self-explanatory).

SQL & Database

8. SQL query: Retrieve employees with salary **above average**.
9. SQL: Fetch employees working in departments with **more than 5 employees**.

Object-Oriented / Programming Language

10. Why Java is widely used? Java vs Python.
11. What did Java 8 introduce?
12. Lambdas and functional interfaces explanation.

13.4 pillars of OOP.

14.Abstract class vs interface.

15.Why Java does not support multiple inheritance.

16.Output of code involving method overriding/inheritance.

17.Questions on HashMap behavior.

Systems / Fundamentals

18.ACID properties discussion.

Resume & Projects

19.*Introduce yourself* & explain your project in detail.

20.Why did you choose your specialization (e.g., AI/ML)?

21.Explain impact and challenges in your project.

Other Conceptual/Behavioral

22.Deep questions on multithreading concepts.

23.OOP vs functional programming.